

? The Question

Why can a field SOH estimate be questioned when it appears to be inconsistent with expectations?

Single-point SOH values from field measurements can show unexpected behavior. Is this a real degradation signal, a measurement artifact, or the result of how the data is collected and interpreted?

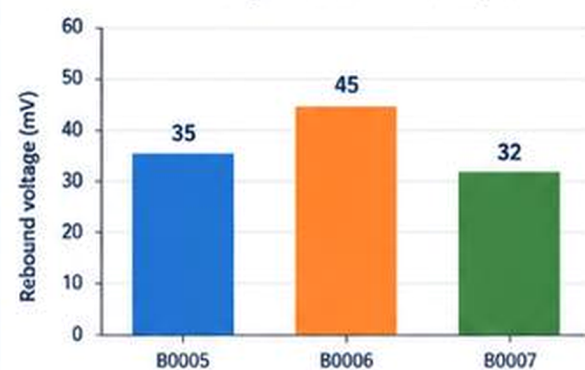
What the Data Shows

The data reveals three key observations:

- 1 Different cells show different baseline levels (cell-to-cell variation).
- 2 Rebound behavior depends on rest time and is not linear across the observed interval.
- 3 Single-point measurements show short-term variation, while longitudinal data provides a clearer trend.

1 Cell-to-Cell Variation

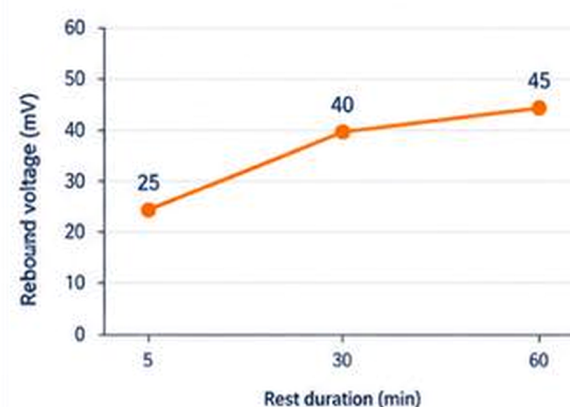
Rebound voltage (mV) at 30 min (example)



→ Different cells have different baseline levels. A single value cannot represent the behavior of the whole system.

2 Rest-Time Dependence

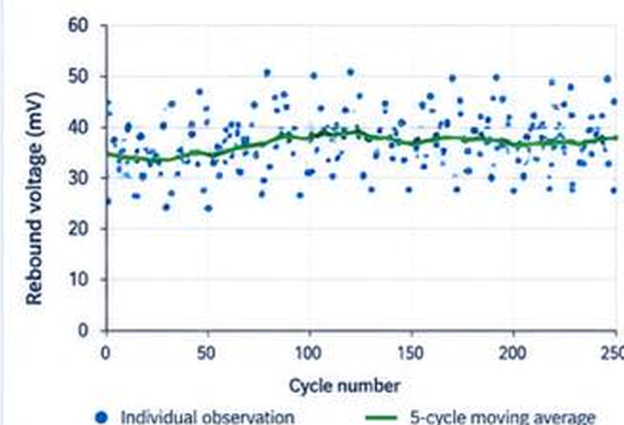
Rebound voltage (mV) vs. rest duration (example)



→ Observed rebound changes with rest duration; the relationship is not well represented by a simple linear extrapolation over these intervals.

3 Single Point vs. Longitudinal View

B0005 rebound voltage (mV): individual vs. 5-cycle moving average



→ Single measurements contain short-term variation; longitudinal data provides additional temporal context.

Engineering Context

- Indirect Observable**
SOH is inferred from measurable signals (e.g., voltage, current, temperature, impedance).
- Cell-Specific Baseline**
Each cell has its own electrochemical behavior and baseline level.
- Rest-Time Dependence**
Relaxation and rebound effects depend on the time after load removal.
- Multiple Uncertainties**
Measurement noise, operating conditions, aging mechanisms and model limitations all affect the result.

Engineering Interpretation

- ✓ A single SOH value is a snapshot, not the full picture.
- ✓ Cell-to-cell variation and rest-time effects can significantly influence the measurement.
- ✓ Longitudinal data helps identify real trends and filter out short-term fluctuations.
- ✓ A more reliable assessment combines multiple signals, temporal context, and uncertainty awareness.

Why It Matters

- Avoid Wrong Decisions**
Reduce the risk of false alarms or missed degradation.
- Improve Accuracy**
Use multi-dimensional data instead of a single point.
- Enable Better Maintenance**
Target the right cells, at the right time, with the right strategy.
- Extend System Lifetime**
More informed decisions lead to safer, more reliable, and longer-lasting BESS.

